
A PIECE OF CHALK, A GREEN LEAF, AND FOUR KINDS OF WHY

Why Are Leaves Green?

A causal descent - and a guide to where explanation becomes evolutionary storytelling

The colour is easy to explain. The history of the colour is not. That gap is not a failure of science. It is an invitation to ask what kind of answer a piece of evidence can actually support.

An evidence-graded essay for upper-secondary students and teachers
17 September 2026

Includes full references, DOI links, and a student reading route

The short answer

Leaves look green because their pigments and internal structure absorb, scatter, and transmit wavelengths unequally; more green light than red or blue commonly escapes toward an observer. But leaves do absorb and use a great deal of green light. Green can penetrate farther through a leaf and a canopy, while excess light can damage photosynthetic machinery. These facts make several evolutionary explanations plausible. They do not yet identify one uniquely established historical reason why the dominant land-plant lineage has a green gap in its absorption spectrum.

The careful answer is therefore layered: the optical *how* is settled; several present-day benefits are strongly supported; the original evolutionary *for what reason* remains active research. A complete causal history could be a perfectly good “reason why” even if no creature intended the result. It would still not automatically be a purpose.

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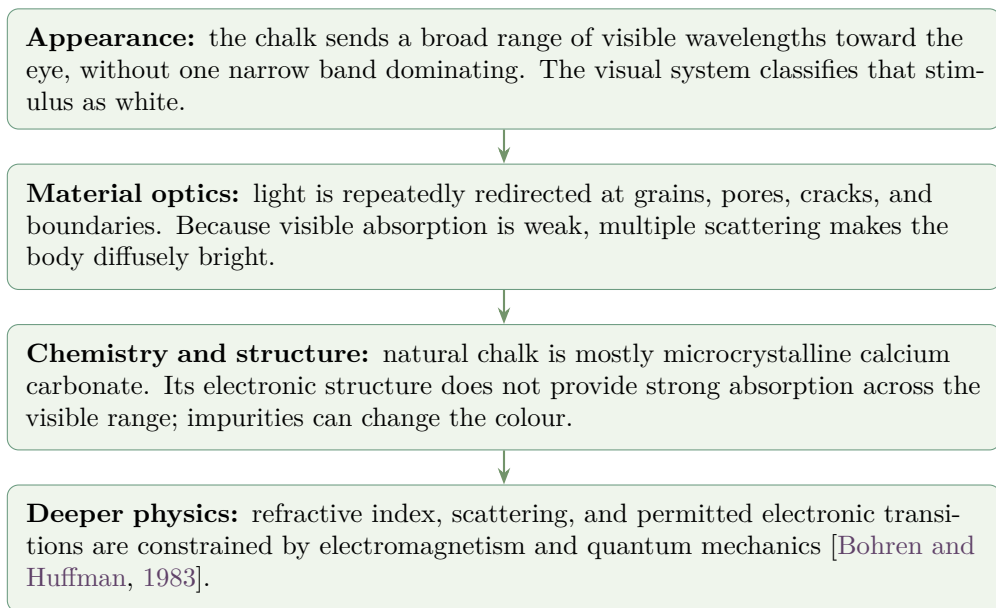
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1 Weinberg's chalk: begin with an ordinary fact

Take a piece of natural chalk. It is white. The sentence feels finished because it is familiar, but familiarity is not explanation. In Chapter 2, “On a Piece of Chalk,” of *Dreams of a Final Theory*, Steven Weinberg used this sort of ordinary object to illustrate an explanatory descent: each answer opens a deeper question, until a fact of everyday appearance is connected to more general physical principles [Weinberg, 1992]. His chapter title also deliberately echoes T. H. Huxley's nineteenth-century lecture on chalk, which descended in another direction - from a lump of chalk into geological and biological history.

The relevant chain can be compressed without pretending that every link is simple:



This is the claim that the links must be mutually compatible. A geological account of how coccoliths accumulated, an optical account of multiple scattering, and a quantum account of electronic transitions answer different questions about the same chalk. Reduction is powerful when it connects levels; it becomes a caricature when it declares the higher levels unreal.

Provenance check

The chalk discussion belongs to *Dreams of a Final Theory* (1992), which matters here because this essay will later argue that knowing where a claim came from is part of knowing what the claim means.

Now put down the chalk and pick up a leaf. Can we make the same descent? Yes - but biology adds a fork that chalk does not. Besides asking how the colour is produced, we can ask how this lineage acquired its pigments, whether the colour does useful work now, and whether natural selection historically favoured it for that work. Those questions are easily collapsed into one sentence beginning “because.” They should not be.

2 A Weinberg-style descent through a leaf

2.1 Level 1: the perceptual event

A leaf is not green all by itself. Colour is a relation among an illuminant, an object, an observer, and a viewing geometry. In ordinary daylight, many healthy leaves return a larger fraction of middle visible wavelengths to the eye than they return of blue and red wavelengths. Human cone responses and neural comparison turn that spectrum into a green experience.

That first sentence already improves on “chlorophyll reflects green.” A pigment molecule mainly *absorbs*; a whole leaf also scatters and transmits. Waxy surfaces can add glare. Air-cell-water interfaces in the mesophyll redirect photons. Some light emerges from the illuminated side as reflectance, some from the far side as transmittance, and some is absorbed. Classic measurements established these wavelength-dependent optical properties across leaves and species [Gates et al., 1965, Woolley, 1971].

2.2 Level 2: pigments and quantum transitions

Chlorophylls *a* and *b* have molecular electronic states that couple strongly to photons in blue and red regions and less strongly to much of the green region. Carotenoids add absorption especially in the blue and also protect the photosynthetic apparatus. The exact spectrum depends on pigments bound inside protein complexes, not on isolated chlorophyll alone.

When an absorbed photon excites a pigment, its energy may be transferred through an antenna complex to a reaction centre. Charge separation then begins an electron-transfer chain that helps produce ATP and reducing power; the Calvin-Benson cycle uses that chemical capacity to incorporate carbon dioxide. Some excitation is lost as heat or fluorescence. Under excess light, regulated nonphotochemical quenching converts additional excitation to heat and helps limit photodamage [Ruban, 2016].

A leaf looks green because green is absorbed less strongly than red and blue over much of the visible range - not because green is unused.

Green photons penetrate deeper before absorption. In strong white light, upper chloroplasts may already be light-saturated while deeper chloroplasts remain able to use additional excitation. Experiments have therefore found conditions in which added green light drives photosynthesis deeper in a leaf, and more effectively at the margin than added red light [Terashima et al., 2009]. A modern review finds that the relative performance of blue, green, and red light changes with intensity, species, anatomy, and the scale being measured [Liu and van Iersel, 2021].

2.3 Level 3: a leaf is not a flat solar panel

“Absorb the maximum possible light” sounds like an obvious design target. It is not the biological target in every environment.

Photosynthesis has bottlenecks. Carbon dioxide supply, temperature, enzyme capacity, nutrient investment, and the rate at which reaction centres can process excitation all matter. At noon, a sun leaf may receive more photons than its biochemistry can productively handle. Absorbing still more can increase heat, reactive chemistry, and repair costs. At the same time, a plant is a three-dimensional canopy: photons captured by a top leaf are unavailable to a shaded leaf below. Weak absorption at one wavelength can distribute light more evenly through tissue and foliage.

This makes an engineering intuition possible: a productive canopy may benefit from not making every upper leaf optically black. Models and measurements in crops suggest that chlorophyll can

sometimes be reduced with little loss of canopy photosynthesis, potentially saving nitrogen or admitting light deeper into the canopy. That does not mean “less chlorophyll is always better,” and one such modelling study did not find a photosynthetic gain from chlorophyll reduction alone under all realistic assumptions [Walker et al., 2018]. It means maximum absorption by the first surface is not identical to maximum carbon gain by the whole plant.

2.4 Level 4: development, inheritance, and history

A leaf becomes green because a developing plant expresses nuclear and plastid genes, builds chloroplasts, synthesizes tetrapyrrole pigments, assembles antenna proteins, and regulates pigment abundance in response to light and stress. Those developmental pathways are inherited modifications of an older photosynthetic lineage.

Here explanation changes character. Physics constrains which molecules can absorb which photons. But it does not select one complete biosphere directly from all physically possible biospheres. Mutations, gene duplications, endosymbioses, extinctions, ecological competition, and environmental change create a branching history. A trait can be retained because it works well enough in a system already built around it. Evolution often remodels an inherited machine; it does not begin each generation with a blank specification sheet.

That is why “why green?” cannot be answered by drawing only the modern solar spectrum beside a modern chlorophyll spectrum. The comparison contains no phylogeny, no extinct alternatives, no fitness measurements, and no record of which variants were available to selection.

3 What is settled, and what is not?

Given that the word *unsettled* can mislead and does not mean that every claim is equally doubtful we rather grade each proposition separately.

Claim	Evidence grade	What the evidence warrants
Healthy leaves commonly look green because their spectral absorption, scattering, and transmission return relatively more green light to an observer.	Established mechanism	Direct optical spectra, pigment chemistry, and tissue optics agree [Gates et al., 1965, Woolley, 1971].
Plants do not use green light.	False	Leaves absorb substantial green light, and green photons can drive photosynthesis, especially deeper in leaves and canopies [Terashima et al., 2009, Liu and van Iersel, 2021].
Reflecting or transmitting some green can reduce uneven excitation and spread light through tissue or a canopy.	Strong present-day support	Physiology and canopy models show real effects, but their size and benefit depend on conditions [Terashima et al., 2009, Walker et al., 2018].
The green gap was originally selected because it prevents light damage.	Plausible, not uniquely shown	Excess light and photoprotection are real [Ruban, 2016]; that does not by itself reconstruct the original selection pressure.
Natural selection tuned pigment spectra to suppress fluctuations or “noise” in energy input.	Supported theoretical model	A noise-cancelling model reproduces several observed antenna spectra [Arp et al., 2020]; fit and plausibility are not a fossil record of selection.
Earlier purple, retinal-based organisms took green light, forcing chlorophyll users into red and blue.	Speculative historical hypothesis	Retinal phototrophy exists and spectral complementarity is interesting, but early phototrophic history is poorly known and direct competition is not established [DasSarma and Schwieterman, 2021].
The Sun’s changing surface temperature explains why modern leaves are green.	Weak as a primary explanation	Solar temperature changed modestly; land plants arose late; atmosphere, water, tissue, and photon accounting reshape the relevant spectrum.
Lineage history and contingency helped constrain the pigment system inherited by land plants.	Very likely in general; details open	Comparative biology strongly supports inheritance and path dependence. The exact sequence of ancient alternatives and selective episodes remains incomplete.

The decisive logical distinction is this:

useful now $\nRightarrow$ originally selected for that use.

A feature can arise as a side effect, become useful later, and then be refined by selection. It can also remain because changing it would require coordinated changes elsewhere. Conversely, a feature once selected can lose its old function when the environment changes. This is why evolutionary biologists have long warned against turning every agreeable present effect into an origin story [Gould and Lewontin, 1979].

4 The Sun, Wien’s law, and the chronology test

The solar idea deserves a fair test because it begins from sound physics. A blackbody’s spectral radiance per unit wavelength has its energy peak at

$$\lambda_{\max} = \frac{b}{T}, \quad b \approx 2.898 \times 10^{-3} \text{ m K}.$$

Using the nominal present solar effective temperature of 5772 K [Prsa et al., 2016],

$$\lambda_{\max} \approx 502 \text{ nm},$$

which lies in the green-cyan part of the visible spectrum. While this calculation is correct the usual conclusion drawn from it of plants oddly rejecting the Sun’s most plentiful photons, is incomplete for four reasons.

4.1 First: a peak depends on what is being counted

Wien’s familiar 502 nm result is the maximum of *energy per unit wavelength*. Photosynthetic chemistry responds to photon number as well as photon energy. If a blackbody spectrum is expressed as photons per unit wavelength, the present solar-temperature peak is near 636 nm, in the red. If it is expressed per unit frequency, the numerical peak moves again. A plotted “peak” is partly a choice of coordinate, not a single privileged packet of sunlight.

4.2 Second: organisms do not receive a bare blackbody

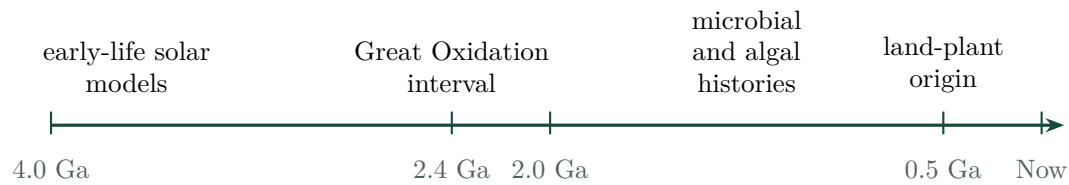
Solar light crosses a stellar atmosphere, Earth’s atmosphere, clouds, water, neighbouring leaves, and finally leaf tissue. Each filter reshapes it. Underwater spectral windows can be especially different from daylight in air. For photosynthesis, the relevant quantity is not simply the Sun’s top-of-atmosphere energy maximum, but the photon spectrum at the organism, modified by what its pigments and tissue can do with those photons [Kiang et al., 2007].

4.3 Third: the young Sun was fainter, but only modestly cooler

Modern solar models place the present effective temperature at 5772 K and show a monotonic rise over the main sequence [Basinger et al., 2024]. Around two billion years ago, a standard-model value is roughly 5700 K (model-dependent and rounded), which would put the wavelength-energy peak near 508 nm. Even a 5600 K young-solar analogue used for much earlier Earth gives about 517 nm [Cnossen et al., 2007]. The shift is real but small compared with the transformations imposed by water, atmosphere, and ecology. The much larger early-solar change was total luminosity: the Archean Sun was roughly 18–25 percent fainter over 3.8–2.5 billion years ago in a recent standard model [Basinger et al., 2024].

4.4 Fourth: the dates do not match a land-plant tale

Oxygenic photosynthesis and the cyanobacterial transformations associated with the Great Oxidation Event belong to a microbial world around 2.4 billion years ago. Land plants arose much later. Molecular-clock analyses place the origin of land plants broadly around 515–470 million years ago [Morris et al., 2018]. Thus the actors two billion years ago were not “submarine plants or fauna migrating onto land.” They were overwhelmingly microbial lineages in aquatic environments. Modern land plants inherited a photosynthetic system whose deep roots long predate their terrestrial radiation.



The chronology does not discount that ancient light environments were irrelevant but rather tells us which organisms and environments a serious hypothesis must address.

5 Purple Earth, green water, and competing histories

The “Purple Earth” proposal is a genuine scientific hypothesis suggesting that simple retinal-based phototrophy may have arisen early and that purple membranes absorbing strongly around green-yellow wavelengths could have been spectrally complementary to chlorophyll systems [DasSarma and Schwieterman, 2021]. Modern microbial rhodopsins show that retinal can harvest light. But several cautions matter:

- A modern biochemical possibility is not evidence that it dominated the early planet.
- Spectral complementarity does not, by itself, demonstrate ancestral competition or prove that chlorophyll was pushed into leftover wavelengths.
- Bacteriorhodopsin-like proton pumping is not the same process as oxygenic carbon-fixing photosynthesis.
- The hypothesis concerns microbes billions of years before land plants, not magenta animals preceding a migration onto land.

Recent work makes the story more interesting rather than simpler. Matsuo et al. [2025] modelled Archean underwater light and argued that iron-related conditions could have made it predominantly green. They combined environmental simulations, engineered cyanobacteria, growth competition, energy-transfer calculations, and phylogenetics to argue that green-absorbing phycobilin antennae gave ancestral cyanobacteria an advantage. This is evidence that early photosynthetic life could evolve *toward* green light in a green-lit habitat. It is not a direct explanation of the green appearance of vascular-plant leaves but it does, however, puncture the idea that chlorophyll-based life universally “cannot use green.”

The live research landscape therefore contains several nonexclusive possibilities:

- 1. Physical and biochemical constraint.** Porphyrin chemistry, reaction-centre energetics, and energy transfer make some pigment arrangements easier or more stable than others.
- 2. Light distribution.** Moderate green absorption sends photons deeper into leaves and canopies, improving the spatial match between excitation and biochemical capacity.
- 3. Protection and regulation.** Avoiding strongly uneven or fluctuating excitation may reduce damage and regulatory cost.
- 4. Ecological spectral partitioning.** Organisms in layered communities can exploit different spectral windows.
- 5. Path dependence.** Once photosystems, pigments, proteins, membranes, and repair pathways coevolved, wholesale replacement may have been difficult even if another design is imaginable.
- 6. Contingency.** Early events could have fixed one workable branch among several, after which selection improved local performance rather than finding a global optimum.

Thus a historically inherited system can be constrained, useful, protective, and later optimized all at once.

6 Four kinds of “why”

Biologists have repeatedly separated proximate causes from evolutionary explanations [Mayr, 1961], and Tinbergen famously distinguished mechanism, development, function, and phylogeny [Tinbergen, 1963]. Dennett, in turn, explored how natural selection can generate design-like “reasons” without a foresighted reasoner [Dennett, 1995]. The fourfold below is an adaptation for this case, in the spirit of Dennett. Its purpose is to prevent a grammatical word - *why* - from doing four scientific jobs at once.

Question	Kind of answer	Leaf answer, with its proper confidence
1. How is it so?	Mechanism	Chlorophyll-protein complexes absorb red and blue more strongly across much of the visible range; internal leaf structure scatters light; relatively more green escapes and reaches the eye. Well established.
2. How did it come to be so?	Development and lineage history	Genes build chloroplasts in each leaf; land plants inherited chlorophyll-based oxygenic photosynthesis through a long algal and cyanobacterial history. Many details of ancient pigment evolution and ecology are reconstructed indirectly. Broad history secure; fine history incomplete.
3. What good does it do now?	Current causal role	Green photons penetrate farther; partial transmission can spread excitation through tissue and canopies; not absorbing every incident photon can limit excess load. Strong, context-dependent evidence.
4. Was it selected for that reason?	Selected-effect function	Perhaps light distribution, damage limitation, energetic stability, noise suppression, ecological partitioning, or a combination increased ancestral reproductive success. Existing evidence does not isolate one original target of selection. Open historical inference.

There is also a fifth question that often hides inside the other four: **Who meant it?** Natural selection needs variation, inheritance, and differential reproduction; it does not need anticipation. If a pigment system left more descendants, there can be a biological reason it was retained without anyone intending green leaves. Dennett’s useful provocation is that there can be reasons before there are reasoners.

But that does not license us to invent the reason. A selected-effect claim is an empirical historical claim. It needs comparative patterns, fitness consequences, plausible ancestral variants, phylogenetic timing, and ideally multiple independent lines of evidence. A trait looking sensible to us is not enough.

7 How researchers turn stories into tests

Evolutionary history rarely arrives as a film. Researchers instead make different traces constrain one another.

7.1 Measure the optical and physiological mechanism

An integrating sphere can measure leaf reflectance and transmittance by wavelength; absorption is inferred from the remainder. Gas-exchange instruments measure carbon uptake. Chlorophyll fluorescence probes how excitation is processed or dissipated. Depth-resolved methods test where

red, green, and blue photons drive chemistry inside a leaf. Varying intensity matters: a colour that is efficient under dim monochromatic light need not have the same marginal value in strong mixed light [Terashima et al., 2009, Liu and van Iersel, 2021].

7.2 Change the system

Mutants, transgenic lines, altered chlorophyll content, and engineered accessory pigments can reveal causal effects. Growth competition under controlled spectra tests whether one pigment state has an advantage in a specified environment. Yet an engineered modern organism is an analogue, not an ancestor. The inference to deep time remains conditional on environmental and phylogenetic evidence [Matsuo et al., 2025].

7.3 Change the scale

A pigment, chloroplast, leaf, canopy, and ecosystem have different optima. Researchers combine radiative-transfer models with measured architecture, enzyme responses, sunflecks, water stress, and nitrogen allocation. A spectrum that seems inefficient in a cuvette can distribute energy effectively through a living canopy [Walker et al., 2018].

7.4 Reconstruct the past

Phylogenies estimate when pigment proteins and pathways diversified. Ancestral-sequence reconstruction can test possible older proteins. Rocks and isotope systems constrain ocean chemistry and atmospheric oxygen. Stellar models constrain luminosity and effective temperature. None alone yields “the reason.” Agreement among independent records makes a historical account harder to fake.

7.5 Make models risk failure

A model that can explain every possible pigment spectrum explains very little. The noise-suppression model of Arp et al. [2020], for example, is interesting because a quantitative principle recovers major absorption features across several photosynthetic systems. Its success is evidence for an organizing principle. It is not yet proof that ancient natural selection literally optimized the model’s objective function. The next questions are discriminating ones: Where does it predict the wrong spectrum? What environmental change should shift the optimum? Can competitors predict the same data with fewer assumptions?

This is close to Deutsch’s demand that a good explanation be difficult to vary while leaving its explanatory work intact [Deutsch, 2011]. “Green protects the plant” is easy to vary: one can substitute cooling, camouflage, ancestral competition, or almost any other agreeable benefit without changing the evidence offered. A stronger account binds its details to risky consequences: alter the spectrum, pigment cost, canopy geometry, or fluctuation statistics, and it should say in advance which absorption profile, growth rate, or damage signature must change.

Question worth keeping open

What observation would distinguish “green is weakly absorbed to spread light” from “green is weakly absorbed because the inherited pigment scaffold made another spectrum difficult”? A good research question asks for evidence that separates explanations, not merely evidence compatible with both.

8 Does everything happen for a reason?

The word *reason* carries at least four loads:

- A. a **cause**: the leaf looks green because of wavelength-dependent absorption and scattering;
- B. a **historical route**: this lineage has chlorophyll because of a sequence of inherited changes and contingencies;
- C. a **biological function**: a feature persisted or was refined because it contributed to reproduction in ancestral populations;
- D. an **intended purpose or meaning**: an agent chose an end, or later interpreters gave an event significance.

A complete causal chain made of lawful interactions, accidents, and happenstance is enough for a reason in senses A and B. It is not automatically a reason in sense C, because selection for an effect must be shown. It is not a reason in sense D unless an intending or interpreting agent enters the story.

Chance and law are not opposites. A mutation can be undirected with respect to an organism's needs while obeying chemistry and physics. A storm can alter a population's future without violating a dynamical law. Natural selection is nonrandom with respect to differential reproductive consequences, yet its raw variation, environments, and sequence of opportunities are historically contingent. Experiments with evolving *E. coli*, for example, show innovations whose later accessibility depended on earlier genetic history [Blount et al., 2008]. An “accident” can become indispensable; later selection can recruit, stabilize, and refine it.

This also sharpens a claim sometimes attributed too broadly to Richard Feynman. Feynman described the atomic understanding of living processes as a powerful assumption, while noting that biology, geology, and astronomy add a historical question: “How did it get that way?” [Feynman et al., 1963]. The laws constrain every step; they do not by themselves specify which lawful history occurred on Earth.

8.1 A Deutsch–Marletto bridge: laws permit; histories select

Constructor theory describes laws partly through possible and impossible transformations, and through systems able to bring possible transformations about repeatedly [Deutsch and Marletto, 2015]. Its counterfactual question for a green leaf is not only how this leaf behaved, but which alternative pigment systems could have been built, copied, and maintained under the same laws.

Marletto argues that accurate self-reproduction can occur under *no-design laws*—laws with no organismal blueprint—if information can be physically copied. In her idealized account, a replicator carries instructions while a vehicle performs construction; together they form a self-reproducer [Marletto, 2015]. In plants, genetic information and cellular machinery help reconstruct chloroplasts and pigments without foresight in the laws.

Constructor theory is a foundational programme, not evidence for the chlorophyll green gap. Its useful distinction is this: physics delimits possible transformations; inheritance makes some repeatable; history determines which survived. It still does not show whether green transmission was selected for light distribution, photoprotection, economy, or retained from a workable ancestor.

So is Earth biology “just” Earth-bound accidents? The word *just* does mischief. A leaf is at once:

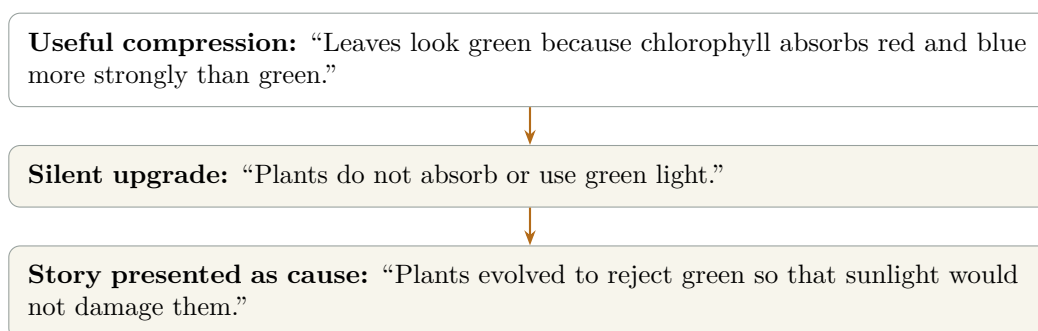
- made possible and constrained by universal physical regularities;
- organized by biochemical and evolutionary processes that have general features;
- one branch of a particular planetary history that could have gone differently.

Meaning is another layer. Biological function can exist without conscious purpose; human meaning can exist without having caused the thing meant. A person may find meaning in a forest even if no forest was intended for that person. Explaining the pigments does not decide what a forest should mean to us. Nor does meaning require us to falsify its history.

9 What school science should do with an unfinished answer

School knowledge must be compressed. No lesson can reproduce every method, caveat, or paper behind every sentence. The danger begins when a useful compression is taught as if it answered a different question.

Consider the familiar progression:



The first is a serviceable first mechanism, though it omits tissue optics. The second is false. The third is a plausible contribution promoted into an established origin story. Students need practice noticing the upgrades.

Research on science learning argues that students should encounter not only conclusions but the authority and accountability practices by which communities make claims answerable to evidence [Ford, 2008]. Work on epistemic cognition likewise asks learners to judge aims, sources, methods, and standards rather than recite that knowledge is merely “tentative” [Chinn et al., 2011, Allchin, 2011]. This need is more urgent when search engines and generative systems can produce fluent, answer-shaped text faster than a reader can inspect its ancestry. Studies show that students across grade levels can struggle to evaluate online claims and sources [McGrew et al., 2018].

Authority is not the enemy. Expertise, peer criticism, replication, instruments, archives, and trusted institutions make modern knowledge possible. The educational alternative to authoritarian certainty is not “believe whatever you like.” It is **accountable trust**: a claim carries the provenance needed to see who made it, by what method, with what scope, and with what degree of confidence.

9.1 A six-line provenance card

For any attractive scientific assertion, a class can record:

Claim	Write the smallest exact proposition. Avoid smuggling mechanism, function, and origin into one “because.”
Kind of why	Observation, mechanism, development, current role, evolutionary history, or selected purpose?
Evidence	Measurement, experiment, comparative pattern, model, phylogeny, geological proxy, or expert synthesis?
Scope	Which species, light intensity, tissue, environment, and time period?
Confidence	Established; strongly supported but nonexclusive; plausible model; or speculative?
Source chain	Textbook or website → review → original paper/data. What does each source actually claim?

The exercise makes one habit routine: *match the strength and kind of the claim to the strength and kind of the evidence.*

9.2 A classroom investigation that preserves the open question

A class can compare a leaf’s reflectance spectrum, chlorophyll extract absorption spectrum, and photosynthetic action spectrum. Even if only published datasets are available, students can ask:

1. Why are the three curves not identical?
2. Which curve concerns appearance, which molecular absorption, and which biological work?
3. Does low reflectance imply high photosynthetic efficiency?
4. What changes at low versus high light, or in a single leaf versus a canopy?
5. Which results show a current effect, and which could establish an evolutionary origin?

The endpoint should be a structured answer whose certainty varies by layer.

10 A reading route for students

The following route moves from accessible orientation to frontier research. Reading an abstract, figures, and discussion critically is often more valuable than struggling through every technical paragraph.

- 1. Begin with the explanatory idea.** Read Weinberg’s “On a Piece of Chalk” in *Dreams of a Final Theory* [Weinberg, 1992]. Ask where he moves from an everyday property to a deeper law, and what is gained or omitted at each level.
- 2. Correct the basic myth.** Read the open review by Liu and van Iersel [2021]. Focus on the sections about penetration depth and intensity. Then examine the title, abstract, and figures of Terashima et al. [2009]. Write one sentence explaining why “green is reflected” does not imply “green is useless.”
- 3. See the question before the latest answers.** Read Nishio [2000]. Notice which observations were secure in 2000 and which evolutionary explanation remained uncertain.
- 4. Move from leaf to planet.** Read Kiang et al. [2007], especially the proposed links among incident photon flux, atmospheric windows, and pigments. Mark every place the authors say *propose*, *may*, or *suggest*.
- 5. Compare two frontier stories.** Contrast the Purple Earth proposal [DasSarma and Schwieterman, 2021] with the green-light Archean reconstruction of Matsuo et al. [2025]. For each, list the modelled evidence, experimental evidence, phylogenetic evidence, and missing direct evidence.
- 6. Audit the chronology.** Use Morris et al. [2018] for land-plant timing and Basinger et al. [2024] for solar evolution. Decide whether the temperature change of the Sun is large enough, and timed appropriately, to serve as the main explanation of green land-plant leaves.
- 7. Learn the philosophy of the question.** Read Mayr on biological causes [Mayr, 1961], Tinbergen on four questions [Tinbergen, 1963], and Feynman’s Section 3-7 online [Feynman et al., 1963]. Then classify every “because” in your own explanation.
- 8. Ask what the laws permit.** Read the abstracts and opening diagrams of Deutsch and Marletto on physical information [Deutsch and Marletto, 2015] and Marletto on life under “no-design laws” [Marletto, 2015]. Translate the leaf problem into three questions: Which pigment systems are physically possible? Which can be reliably reconstructed? Which were historically selected? Keep those answers separate.
- 9. Learn to inspect knowledge itself.** Read Allchin’s account of “whole science” [Allchin, 2011] or Ford on authority and accountability [Ford, 2008]. Apply the provenance card to one science-textbook paragraph.

Question worth keeping open

If an extraterrestrial biosphere under a Sun-like star were black, purple, or orange, which parts of our explanation would travel with us as physics, which as evolutionary principles, and which would remain peculiar facts about Earth’s history?

11 Coda: the leaf after the explanation

Why is the leaf green? Because certain photons are absorbed and others escape. Because molecular orbitals and protein environments make some transitions likely. Because a folded tissue scatters light. Because chloroplasts were built from instructions inherited through an immense lineage. Because green photons can reach places red photons have already saturated. Perhaps because avoiding or redistributing excess excitation paid an ancient dividend. Perhaps also because evolution began from a workable pigment system and could only move through neighbouring possibilities. Some of these sentences report measurements. Some summarize models. Some are disciplined historical conjectures.

If one day those conjectures converge into a well-supported causal chain containing accidents, constraints, and selections, we will have a reason why in the historical sense. No intention is required. If differential survival preserved parts of the chain because of what they did, we may also speak carefully of biological function. Purpose in the conscious sense would still be another question; meaning, something minds and cultures can make, another again.

Physics gives biology no exemption from law. Biology gives explanation something physics often lacks: inherited organization and a path through time. Feynman's point was not that life is less lawful, but that its account must ask how it got that way. And his closing counsel remains apt: "It does not do harm to the mystery to know a little about it" [Feynman et al., 1963].

In an age of easy answers, perhaps the hardest school lesson is also the most hopeful: knowledge is not weakened when we show its joints. A statement becomes more trustworthy when a learner can see which part is observed, which inferred, which modelled, which inherited from an authority, and which still waits for a better test. Chalk can teach the descent toward law. A leaf can teach why the descent is not the whole journey.

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